E202 Towards a Modern Atomic Model

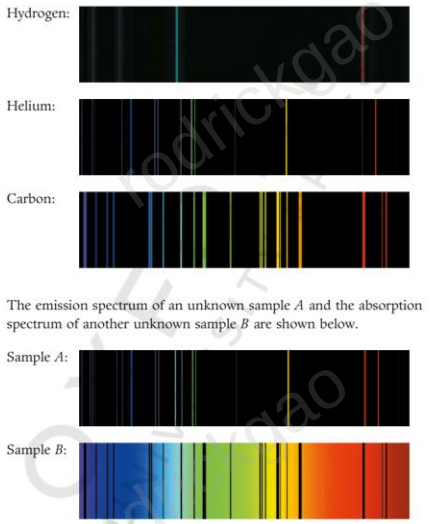
E2.2.1 Rutherford's model and its limitation

Scattering supports an atom that is mostly empty space, with positive charge and almost all mass concentrated in a tiny nucleus. A classically orbiting electron accelerates and should radiate energy, making stable atoms difficult to explain. Bohr's discrete stationary states address this problem; classical orbits alone do not provide a complete atomic model.

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E2.2.2 Identifying elements from spectral lines

Compare an unknown sample's line wavelengths with known elemental patterns. Bright emission and dark absorption lines of the same element occur at corresponding wavelengths. Match the positions of the whole set rather than only line count or brightness. In the illustrated example, sample A's emission includes hydrogen and helium, while sample B's absorption matches carbon.



Match known elements and unknown samples at the same spectral-line wavelengths.

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E2.2.3 · E2.2.4 Stationary states and angular momentum

Bohr's hydrogen model allows only stationary orbits with quantised angular momentum. Electrons do not radiate in stationary states; transitions absorb or emit photons with energies equal to level differences. The quantum number is a positive integer, not a continuous variable. This hydrogen model cannot be extended unchanged to all many-electron atoms.

$$m_e v_n r_n = \frac{nh}{2\pi}$$

$$L = \frac{nh}{2\pi}, n = 1,2,3, \ldots$$

Bohr hydrogen model: m_{e}: electron mass (kg); v_{n}: orbital speed (m s⁻¹); r_{n}: radius (m); n: positive integer; h: Planck constant (J s).

Bohr hydrogen model: electric attraction supplies centripetal force for circular orbits. Only discrete stationary orbits are allowed; an electron in one such orbit does not radiate and has constant energy. Transitions between orbits involve absorption or emission of a photon with the energy difference. Orbital angular momentum L (J s) is quantized; h = 6.63×10^{-34} J s.

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E2.2.5 · E2.2.6 · E2.2.7 · E2.2.8 Hydrogen orbits, levels and energy changes

E2.2.5 Orbital radii in hydrogen

In Bohr's hydrogen model, the radius of state n is proportional to n squared, so doubling the quantum number quadruples radius. Only positive integer n values are allowed and n = 1 is the ground state. Distinguish this model's radius relation from energy levels approaching zero as n increases.

$$r_n = n^2 r_1$$

Bohr hydrogen atom: r_{n}: radius of orbit n; r_1: ground-state radius; n = 1, 2, 3, … .

E2.2.6 Hydrogen energy levels

With zero energy assigned to a stationary electron at infinity, bound hydrogen levels are negative. The ground state n = 1 has the lowest energy and n = 2 is the first excited state. Larger n gives levels closer

to zero and more closely spaced. Negative energy means binding; required transition energy is final minus initial energy.

$$E_n = \frac{E_1}{n^2} = \frac{-13.6\ \mathrm{eV}}{n^2}$$

Hydrogen atom; zero energy is a stationary free electron infinitely far from the nucleus. n: positive integer; E_1 = −13.6 eV.

E2.2.7 Excitation energy

Excitation raises an electron from a lower to a higher bound level, requiring final minus initial energy. This card's expression starts from the ground state; use the actual initial level if the atom is already excited. For a one-photon bound-state transition, photon energy must match an allowed level difference.

$$E_{\text{exc}} = E_n - E_1$$

Excitation from ground state to bound level n > 1. Energy difference is positive; use consistent energy units.

E2.2.8 Ionisation energy

Ionisation removes a bound electron to infinity. At threshold, final energy is zero, so the minimum required energy is the negative of the initial bound energy. Ground-state hydrogen requires 13.6 eV; excited states require less. Energy above threshold can become free-electron kinetic energy rather than remaining in a bound level.

$$E_{\text{ion}} = E_\infty - E_1 = 13.6\ \text{eV}$$

Hydrogen atom initially in its ground state; E∞ = 0. From excited level n, ionization requires 13.6/n² eV.

State	Quantum number	Energy level	Radius of orbit
ground state	$n = 1$	$E_1 = -13.6$ eV	$R_1 = 5.29 \times 10^{-11}$ m
1st excited state	$n = 2$	$E_2 = -\frac{13.6}{2^2} = -3.40$ eV	$R_2 = (2)^2(5.29 \times 10^{-11}) = 2.12 \times 10^{-10}$ m
2nd excited state	$n = 3$	$E_3 = -\frac{13.6}{3^2} = -1.51$ eV	$R_3 = (3)^2(5.29 \times 10^{-11}) = 4.76 \times 10^{-10}$ m
3rd excited state	$n = 4$	$E_4 = -\frac{13.6}{4^2} = -0.85$ eV	$R_4 = (4)^2(5.29 \times 10^{-11}) = 8.46 \times 10^{-10}$ m

Table 2.2b *Energy levels and radii of orbits in different states.*

As quantum number n increases, bound-state energy approaches zero from below and orbit radius increases. The energy spacing becomes smaller.

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E2.2.9 · E2.2.10 Energy transitions and spectral wavelengths

An electron dropping to a lower level emits a photon equal in energy to the level difference. A larger difference means higher frequency and shorter wavelength. Use hydrogen energy levels in the correct order to obtain positive photon energy. With SI Planck constant, convert level differences from eV to J first.

$$hf = \frac{hc}{\lambda} = E_a - E_b$$

$$\frac{1}{\lambda} = \frac{13.6\ \text{eV}}{hc}\left(\frac{1}{b^2} - \frac{1}{a^2}\right)$$

Emission from higher level a to lower level b, a > b. Photon energy is positive. Convert eV to J when using h in J s; λ is vacuum wavelength.

Hydrogen emission a > b. Convert 13.6 eV to joules if h is in J s and c in m s⁻¹; then λ is in metres.

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E2.2.11 Photon momentum

A photon has no rest mass but carries energy and momentum. Its momentum magnitude is inversely proportional to vacuum wavelength and directed along propagation. Shorter-wavelength photons have greater momentum and can transfer it through absorption or reflection. Substituting zero rest mass into the ordinary non-relativistic mass-times-velocity formula does not give photon momentum.

$$p = \frac{h}{\lambda}$$

Photon momentum p (kg m s^{-1}); h = 6.63 × 10^{-34} J s; λ: vacuum wavelength (m).

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E2.2.12 The de Broglie wavelength

Matter particles have wavelengths associated with momentum: larger momentum gives shorter wavelength. Replace momentum by mass times speed only in the non-relativistic regime. Electron wavelengths can show observable wave effects at suitable scales, while macroscopic wavelengths are tiny. Accelerating electrons raises momentum, shortens wavelength and improves the wavelength-related diffraction limit.

$$\lambda = \frac{h}{p} \approx \frac{h}{mv}$$

de Broglie wavelength λ (m); momentum p (kg m s^{-1}); h = 6.63 × 10^{-34} J s. p ≈ mv only at non-relativistic speed.

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